Student Name (in English): _____ Student ID: _____

Question1: True / False (0.25 for each) (5) Minutes [1.5] / 1.5 Marks

#	Question	T/F
1	Radix sort is in-place	F
2	Counting sort is stable	T
3	For all weighted graphs and all vertices s and t, Bellman-Ford starting at s will always return a shortest path to t.	F
4	If all edges in a graph have distinct weights, then the shortest path between two vertices is unique.	T
5	To sort n integers in the range from 1 to n^2 , a good implementation of radix sort is asymptotically faster in the worst case than any comparison sort.	T
6	RADIX SORT does not work correctly (i.e., does not produce the correct output) if we sort each individual digit using INSERTION SORT instead of COUNTING SORT	F

Question 2: **Sorting** (10) Minutes [ ] / 2 Marks

1- Sort using the count method the list A={ 5,7,5,2,1,1}

A

1	2	3	4	5	6
5	7	5	2	1	1

B

1	2	3	4	5	6
1	1	2	5	5	7

C

1	2	3	4	5	6	7
2	1	0	0	2	0	1

C`

1	2	3	4	5	6	7
2	3	3	3	5	5	6

1	2	3	4	5	6	7
~~2~~	~~3~~	3	3	~~5~~	5	~~6~~

0 2 4 ~~5~~

3

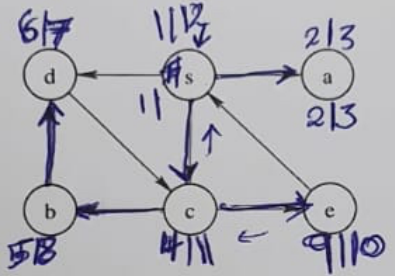
Final C` =

1	2	3	4	5	6	7
0	2	3	3	3	5	5

Student Name (in English): _____ Student ID: _____

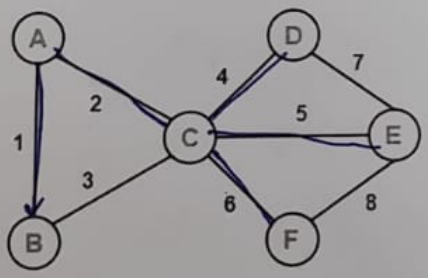
Question 3: Graph **(20) Minutes [4] / 4 Marks**

1- Show how Depth-first-search work on the following graph. Start with the vertex S and use alphabetical order to break ties. Show your work in the tables below. **[2 marks]**



Vertex	A	B	C	D	E	S
d	2	5	4	6	9	1
f	3	8	11	7	10	12
π	S	C	S	b	C	

2- Use undirected-weighted graph below to find a minimum spanning tree (MST) by using Prim's algorithms, draw the resulting minimum spanning tree and specify the value of MCST **[2 marks]**



Step 1 (A) adj[A] = {B, C}
Q = {B, C, D, E, F} π[B] = A π[C] = A
key[B] = 1 key[C] = 2

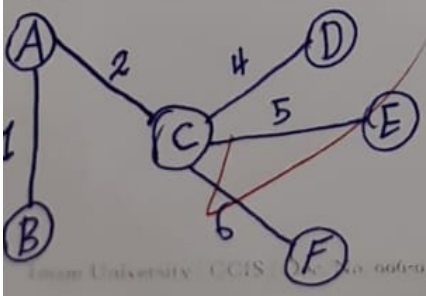
Step 2 (B) adj[B] = {C} ignore

Step 3 (C) adj[C] = {D, E, F}
π[D] = C, π[E] = C, π[F] = C
key[D] = 4, key[E] = 5, key[F] = 6

Step 4 (D) adj[D] = {E} ignore

MCST = 10 + 8 = 18

Nodes	A	B	C	D	E	F
key	0	1	2	4	5	8
π		A	A	C	C	C
Q	A	B	C	D	E	F



Good luck